

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP

#XUV♥
HYDROCARBONS

LECTURE-03

CHEMISTRY

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Topics To Covered

- 1** Chemical Properties of Alkanes.



JK BOSE

WALLAH

TARGET 90

GET READY FOR
10TH AUGUST

THE **REAL** THING IS COMING!

Revision

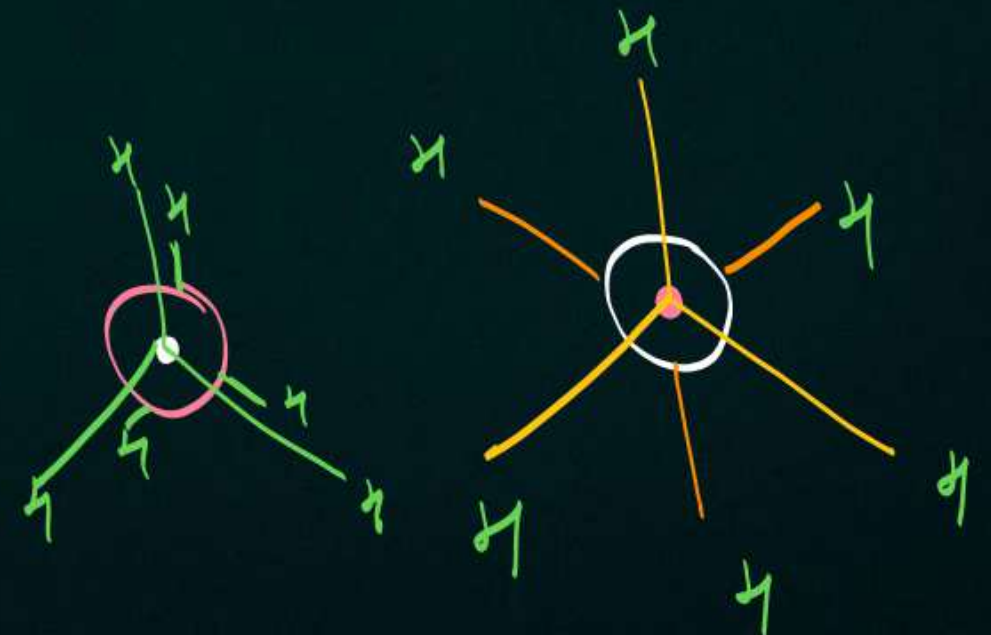
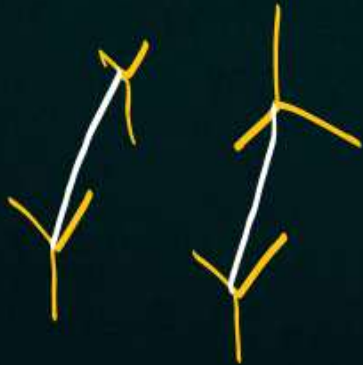
Conformations of Ethane.



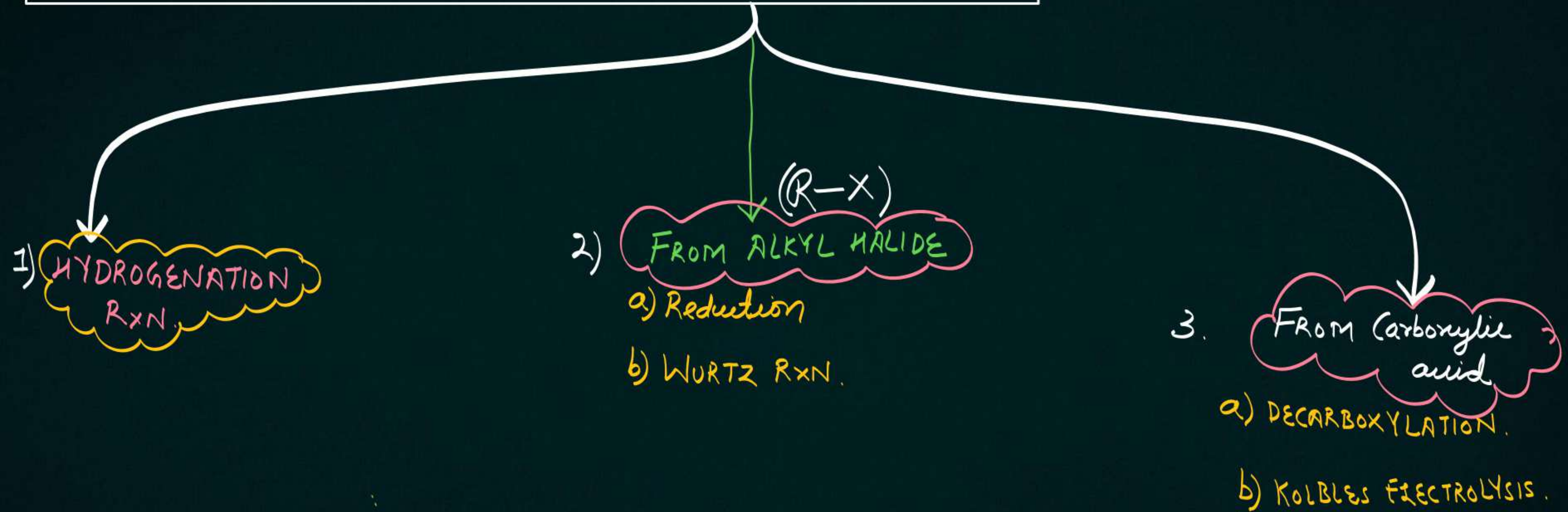
Represent $\Rightarrow$

SAWHORSE

NEWMANN



METHODS OF PREPARATION OF ALKANES.

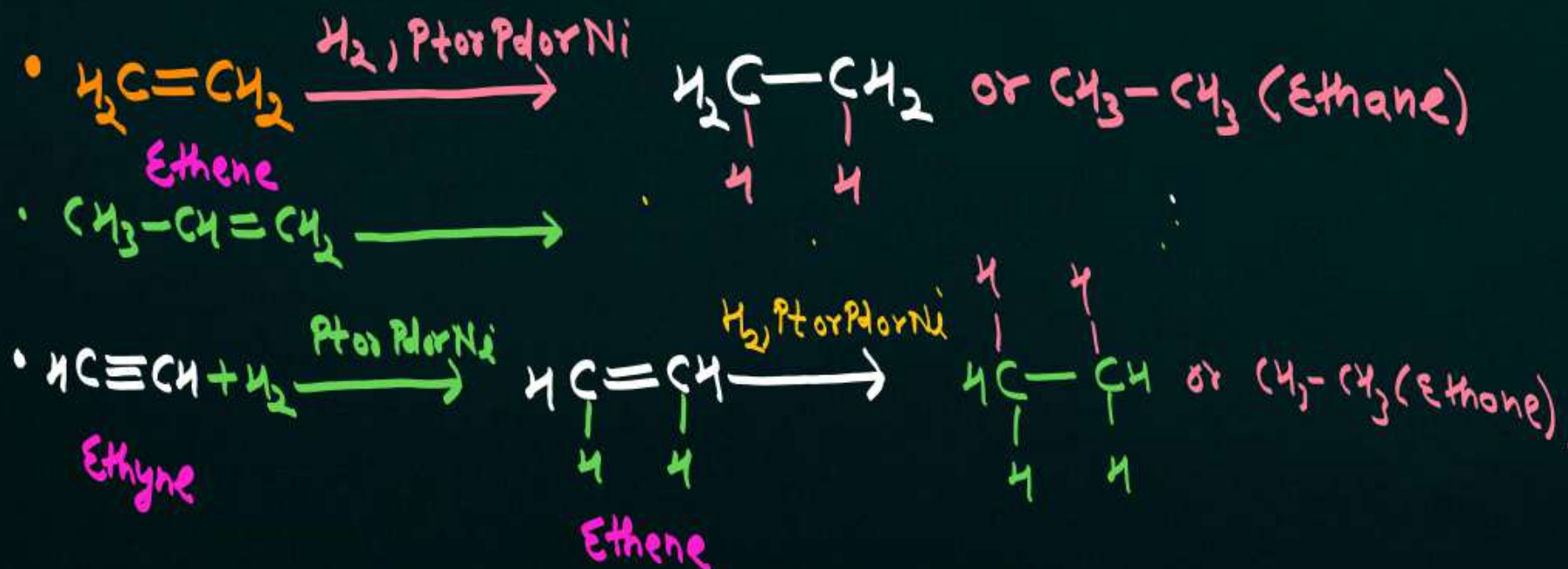


Preparation of Alkanes



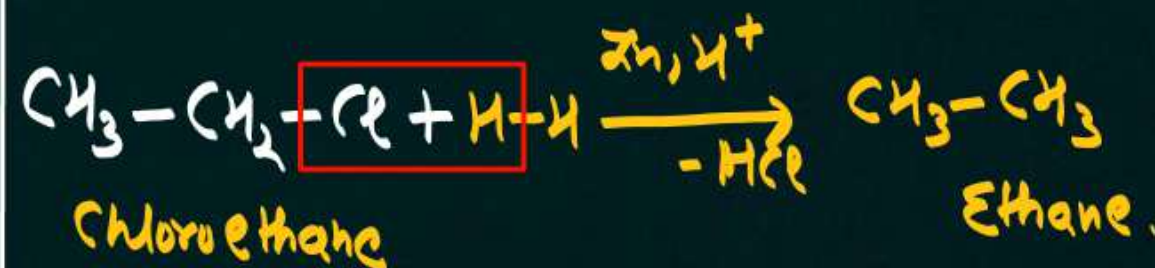
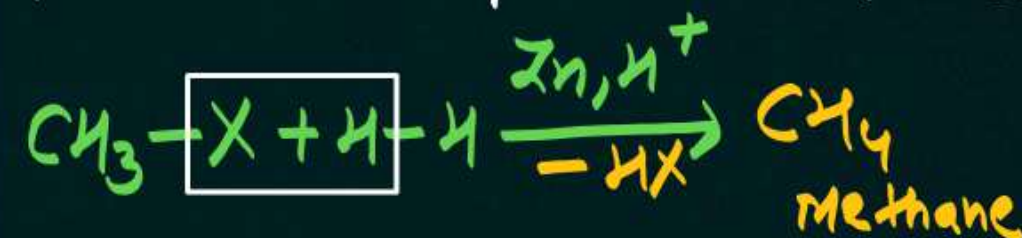
1. Hydrogenation

- Alkenes and alkynes react with hydrogen gas (H_2) in the presence of catalysts like Pt, Pd, or Ni to give alkanes.
- This reaction is called hydrogenation.



2. From alkyl halides

a) REDUCTION $\rightarrow$ Alkyl halides on reaction with H_2 in presence of Zn & acid produces respective alkane.



b) WURTZ REACTION. Alkyl halides on reaction with sodium metal in presence of solvent dry ether produce respective alkane.



- Methane cannot be prepared.
- Only symmetrical alkanes gives good yield.
- If two different alkyl halides are used then mixture of alkanes are formed.

Preparation of Alkanes

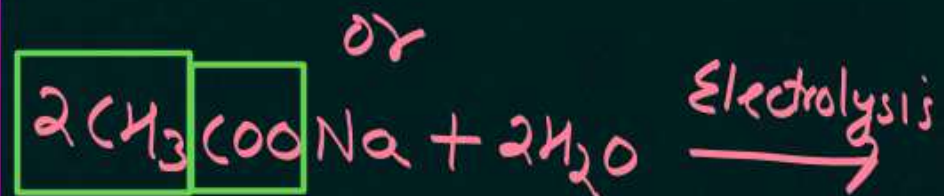
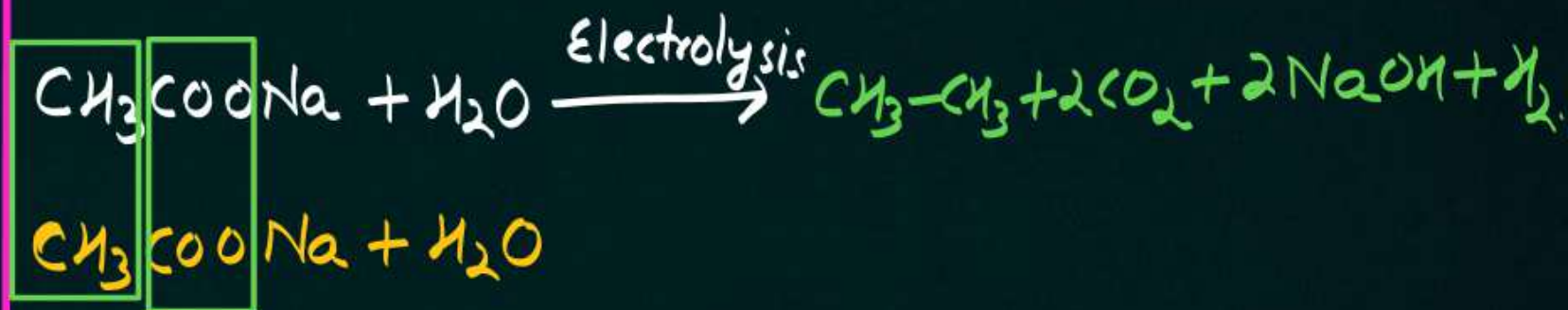
3. From Carboxylic Acids $(-\text{COOH})$

a. Decarboxylation $\rightarrow$ Salt of carboxylic acid on reaction with Soda lime (NaOH ₍₃₎ + CaO ₍₁₎) produces respective alkane.

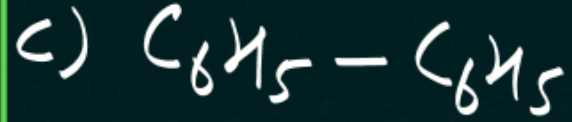
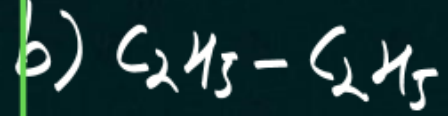
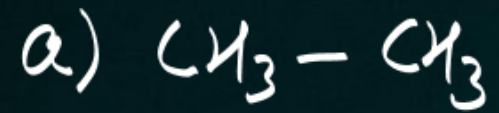
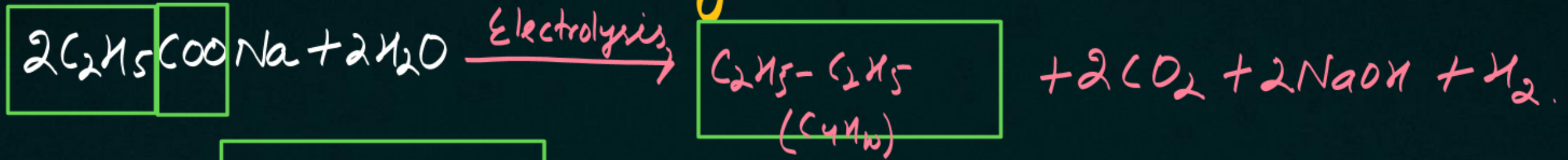


b) KOLBE'S ELECTROLYSIS $\rightarrow$

Salt of Carboxylic acid on electrolysis with water produces respective alkane



Q. Write the Product of following Reaction $\Rightarrow$





Preparation of Alkanes

Kolbe's electrolytic method

Physical Properties of Alkanes

(a) Physical State

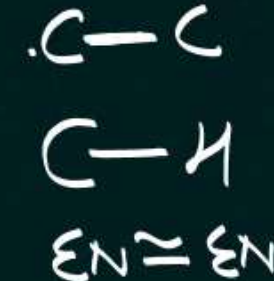
Alkanes have weak van der Waals' forces.

$C_1 - C_4 \rightarrow$ gases

$C_5 - C_{17} \rightarrow$ liquids

$C_{18} \rightarrow$ waxy solids

Gas (1-4) Liquid (5-17) Solid (18+)



(b) Dipole Moment

Alkanes have only C—C and C—H bonds with very small electronegativity difference.

Hence, they are non-polar and have zero dipole moment.

(c) Solubility

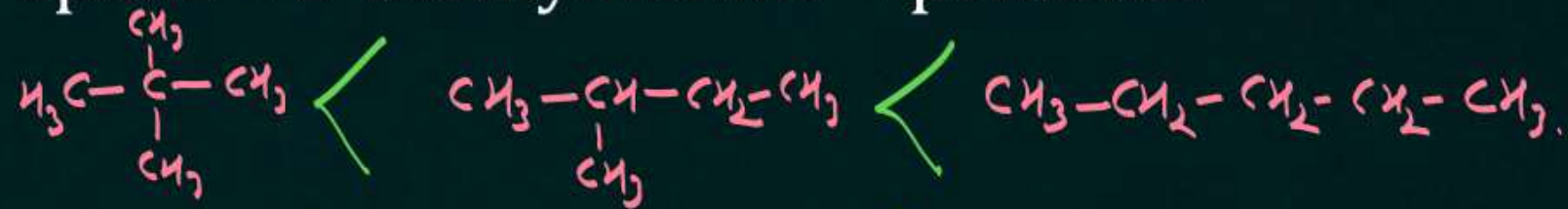
Being non-polar, alkanes are insoluble in water but soluble in organic solvents like alcohol, ether, benzene. Solubility decreases as molecular weight increases.

Physical Properties of Alkanes

(d) Boiling Point

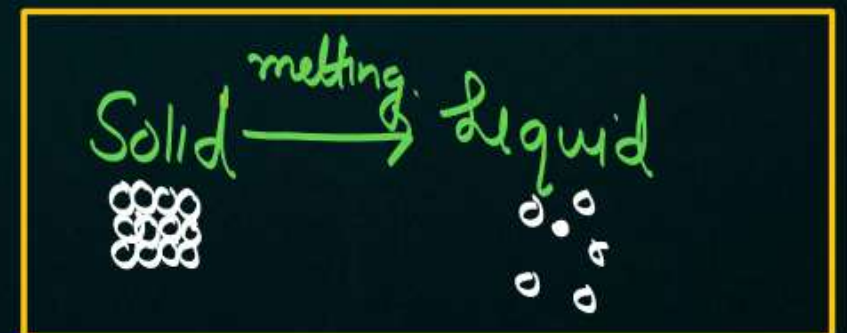
$B.P \propto \text{Mol. wt} \propto \text{Inter-molecular FWA}$
 If MW same (Isomer) $\rightarrow B.P \propto \frac{1}{\text{Branching}}$ $\propto$ Surface Area.

- Boiling point increases with molecular mass (more van der Waals' forces).
- (Isomers) Branching decreases boiling point (smaller surface area $\rightarrow$ weaker forces).
- Example:** 2, 2-dimethylpropane < 2-methylbutane < pentane.



(e) Melting Point & Density

- Melting point rises with molecular weight. ($M.P \propto M.W$)
- Alkanes with even carbons have higher melting points (better packing). ($M.P \propto \text{Symmetry}$)
- Alkanes are lighter than water, and density increases slightly with molecular mass.



Q. Identify the correct order of Boiling Point. (Hint. BPd MW).



QUESTION



Soda lime is a mixture of (3:1)

- 1 $\text{NaOH} + \text{Ca}(\text{OH})_2$
- 2 $\text{NaOH} + \text{Mg}(\text{OH})_3$
- 3 $\text{NaOH} + \text{CaO}$
- 4 $\text{NaOH} + \text{MgO}$



QUESTION

Among the following, which hydrocarbon is not produced by **Wurtz reaction**?

- 1** Methane
- 2** Ethane
- 3** Propane
- 4** All given options can be prepared

NOTE $\Rightarrow$ ORGANIC CHEMISTRY is Just like Maths.
We have to practice it.

REVISE KARNA LIKH KE

Thank
You

*Take
Care*